Wenlong Xiong

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Experience

Robinhood, Senior Software Engineer

Traffic Infrastructure — Feb 2023 - Present

- Implemented an Egress Gateway using Envoy for egress filtering, and led a migration effort to secure outbound traffic across 100+ services and multiple protocols (https, ssh, ftp, websocket).
- Maintained and supported gRPC Golang / Python libraries, gRPCLB / XDS loadbalancers, and performed technical risk assessment of existing Traffic stack.
- Designed and built the Websocket Gateway powering all notifications for Robinhood Legend.
- Spearheading Robinhood's migration to Istio Service Mesh.

Kafka and Streaming Platform — Jun 2021 - Feb 2023

- Helped design and implement RH's standard Golang Kafka library, which included features like L7 client auth, deadletter queues, default observability, and multicluster Kafka support. Adopted by majority of existing and new backend services (80+).
- Led RH's Kafka reliability initiative and migrated all critical order path services to multicluster kafka clients. This eliminated single Kafka clusters from being SPOFs for trading flow, and also unlocked zero-downtime maintenance for more frequent cluster upgrades.
- Designed and implemented a Kafka consumer proxy sidecar in Java. This decoupled application logic from client libraries, reduced version upgrade risk in a monorepo, and greatly decreased consumer rebalance issues on the trading path; while matching existing throughput and latency requirements (∼1-200k msgs/sec per topic, ∼1s max consumer lag from a cold start).
- Defined 2 year roadmap for kafka infrastructure given future scaling & staffing constraints. Documented Build vs Buy decision after evaluating vendors (Confluent Cloud, MSK) & open source (Strimzi, etc). Built proof of concept for Kafka on kubernetes and defined migration path, as well as medium-term cost savings like infrastructure rightsizing, saving 2.5MM/Y.
- Worked with team to authn/z all Kafka traffic, migrating ∼100 services across a dozens of clusters.
- Helped implement a CDC system using Debezium & Flink for event-driven applications.

Data Infrastructure — Feb 2020 - Jun 2021

- Oversaw RH's Spark/Hadoop multicluster (1 to 15) migration, and built initial version of job routing system. This enabled tenant isolation and higher SLOs, unblocking RH's first data-intensive product (IPO Access), and met compliance requirements for IPO (financial data access controls).
- Developed a framework for managing ordering & lifecycle of Kubernetes sidecars, now used in all RH production kubernetes services (100s of namespaces, dozens of clusters).
- Bootstrapped Robinhood's Data Engineering team & interviewed first ∼10 initial hires.
- Drove Airflow 1.x to 1.10 upgrade and kubernetes migration for hundreds of DAGs, dozens of services.
- Derisked and started kubernetes migration for production Presto/Trino clusters.

Microsoft (Azure Stack), Software Engineering Intern — Jun 2019 - Sep 2019

- Compute infrastructure (VM provisioning) team for hybrid cloud. Enabled IMDS (Instance Metadata Service) in Azure Stack VMs, and extended existing Azure Stack .NET services to support this newly introduced component.

Bloomberg, Software Engineering Intern — Jun 2018 - Sep 2018

- Wrote an event ingestion service in Python that performed real time filtering and redaction on messages sent using Instant Bloomberg Chat.

Skills

Languages: Golang, Python, Java, Bash

Technologies: Kubernetes, Envoy, Istio, Kafka, gRPC, Terraform, SaltStack, Airflow, Spark, Hadoop, Presto/Trino

Links

Robinhood Legend, Introducing Robinhood Legend — 2024

Current Kafka Summit, Kafkaproxy: Decoupling Kafka Consumer Logic from Application Business Logic — 2023

Current Kafka Summit, Dead Letter Queues for Kafka Consumers — 2022

Education

University of California, Los Angeles

M.S. Computer Science; GPA: 3.93 — Sep 2018 - Dec 2019

B.S. Computer Science; GPA: 3.76 (*Cum Laude*) — Sep 2014 - Jun 2018